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(54) **BURROWING ANIMAL TRAPS**

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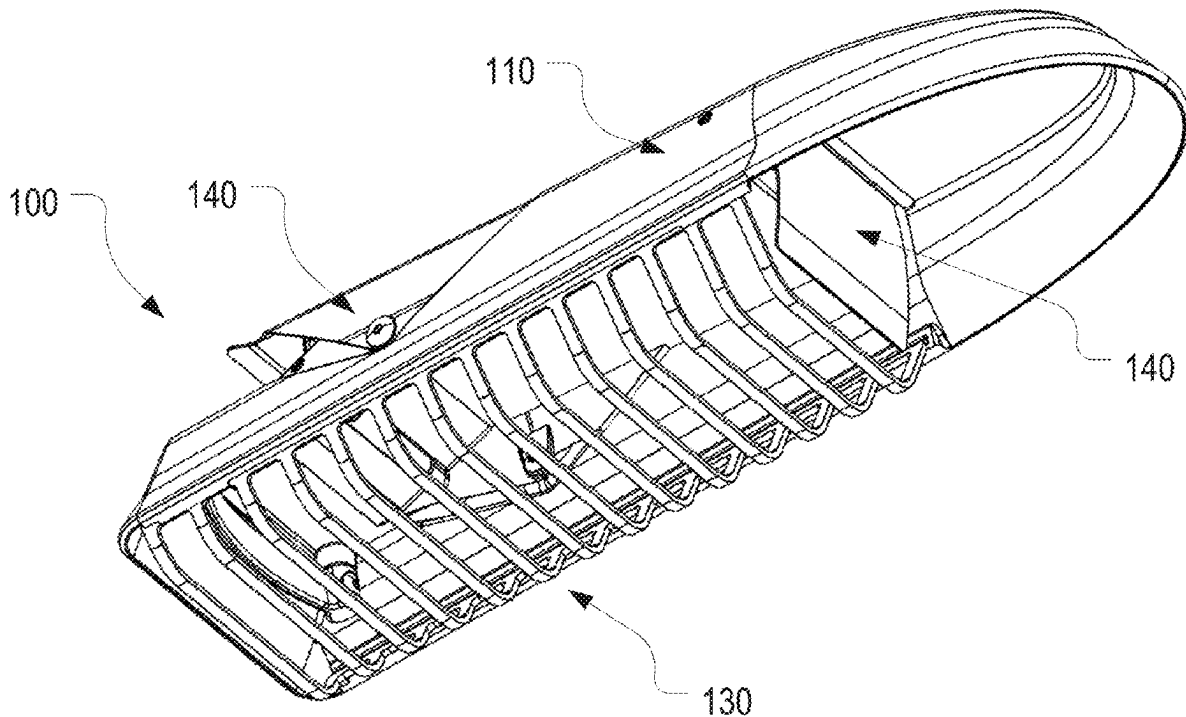
(57) **ABSTRACT**

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Related U.S. Application Data

(60) Provisional application No. 63/563,552, filed on Mar. 11, 2024.

A burrowing animal trap is provided and is designed to enclose the entrance to a burrow and appear to the animal as an extension of the burrow. As a result, the animal will be likely to enter the animal trap under the mistaken belief that it is exiting the burrow. The burrowing animal trap has a capture mechanism that includes a pivoting trap door.



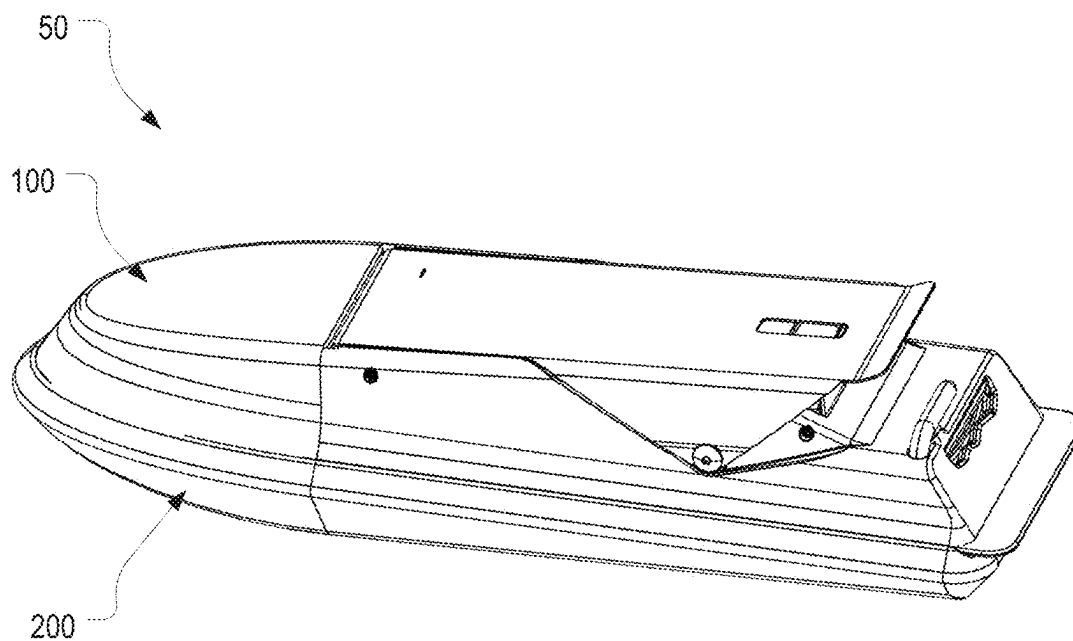


FIG. 1A

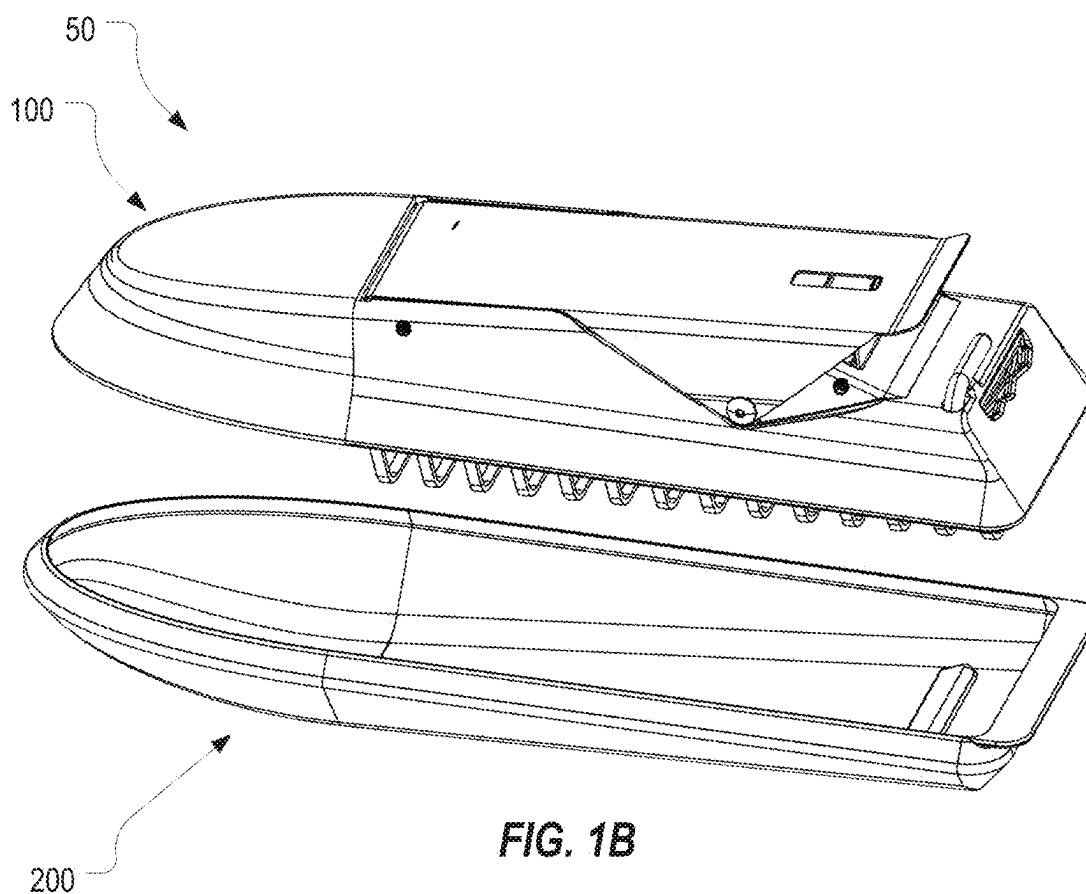


FIG. 1B

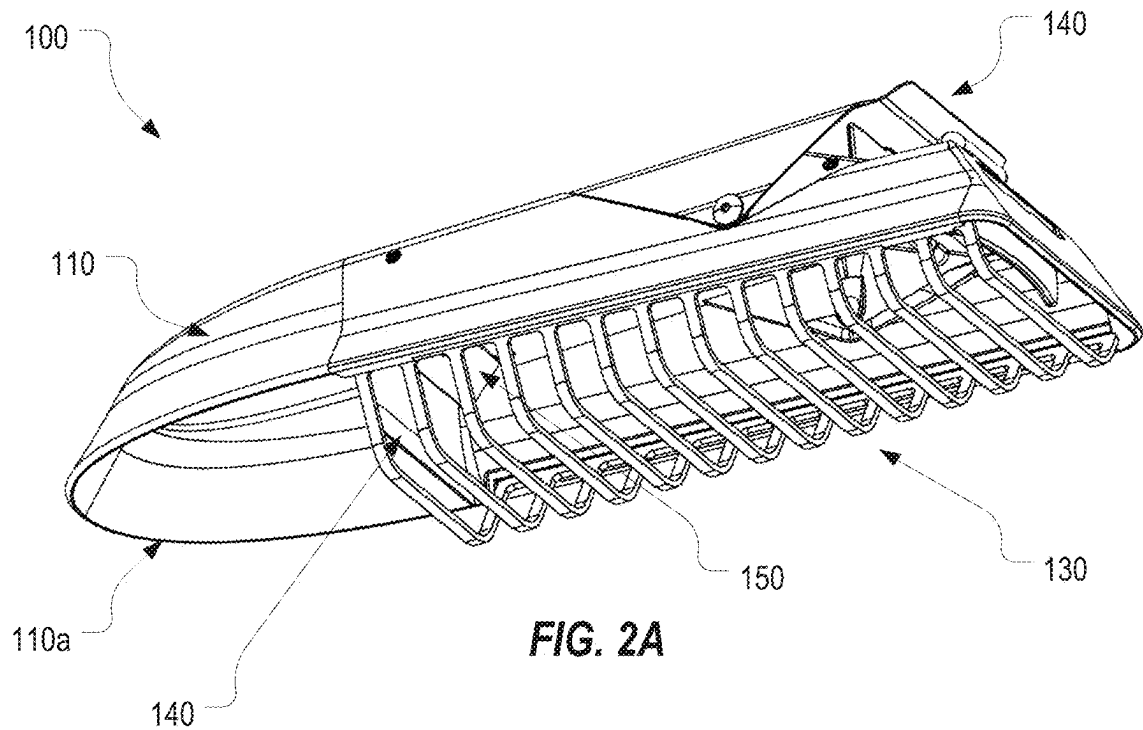


FIG. 2A

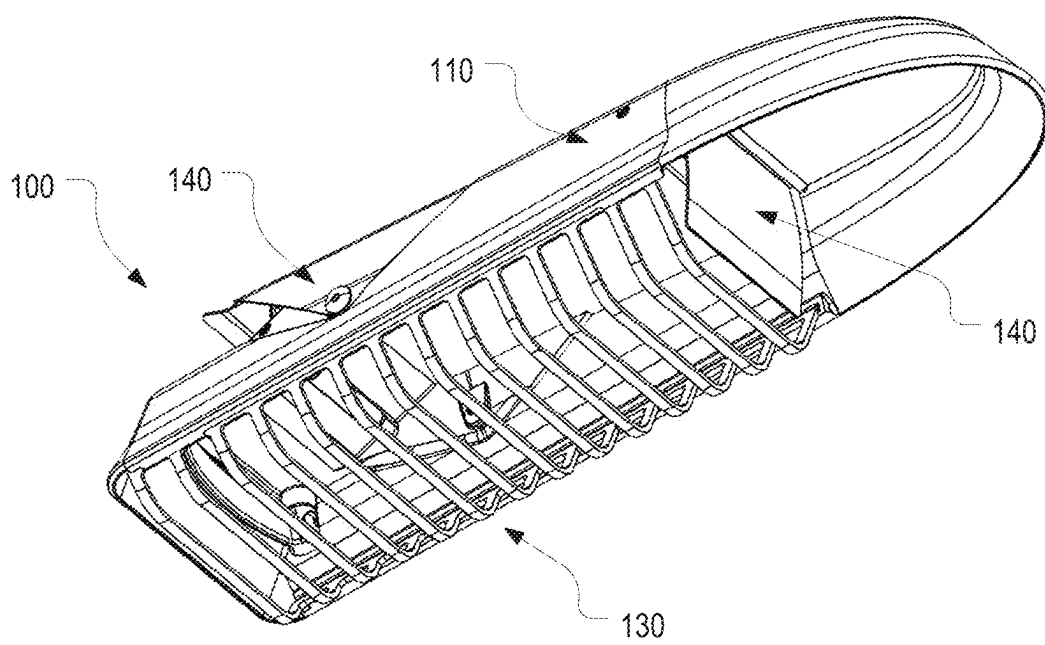


FIG. 2B

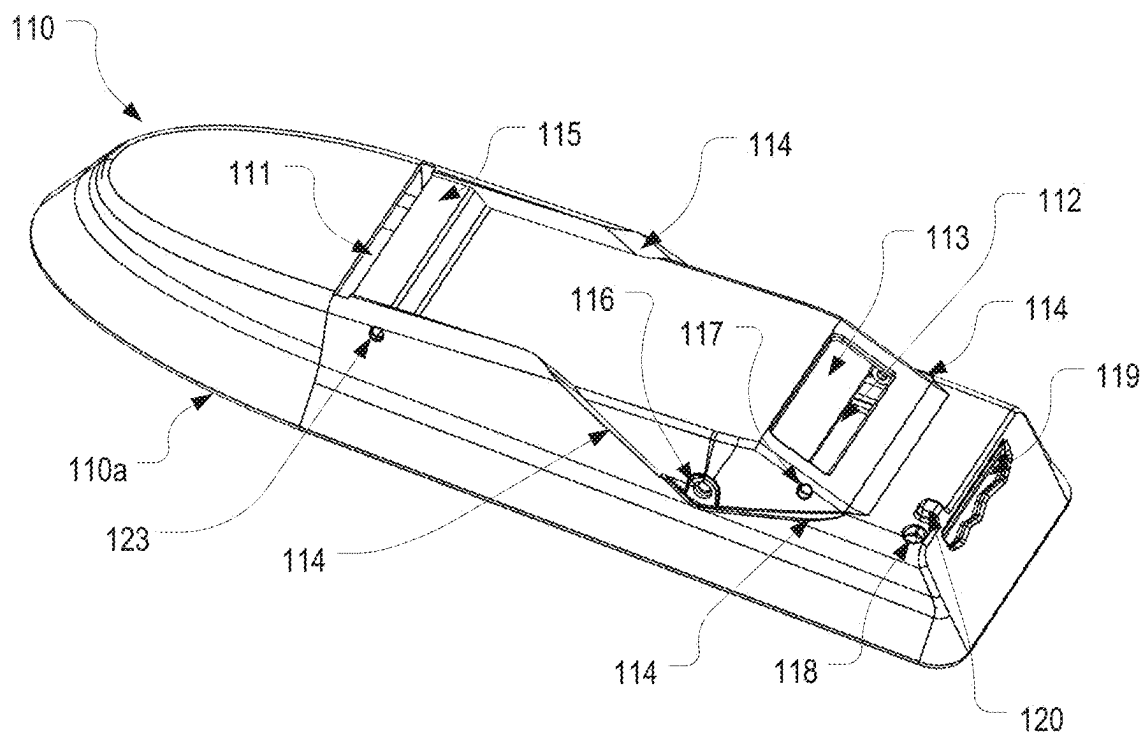


FIG. 3A

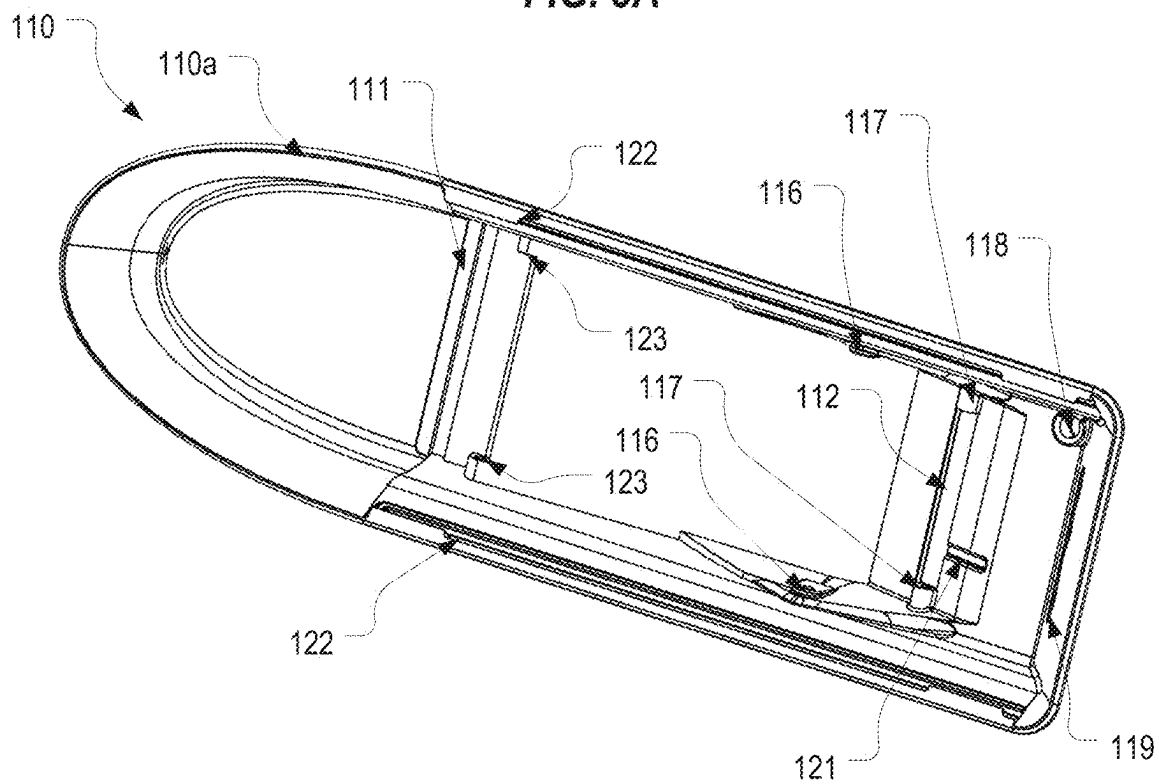


FIG. 3B

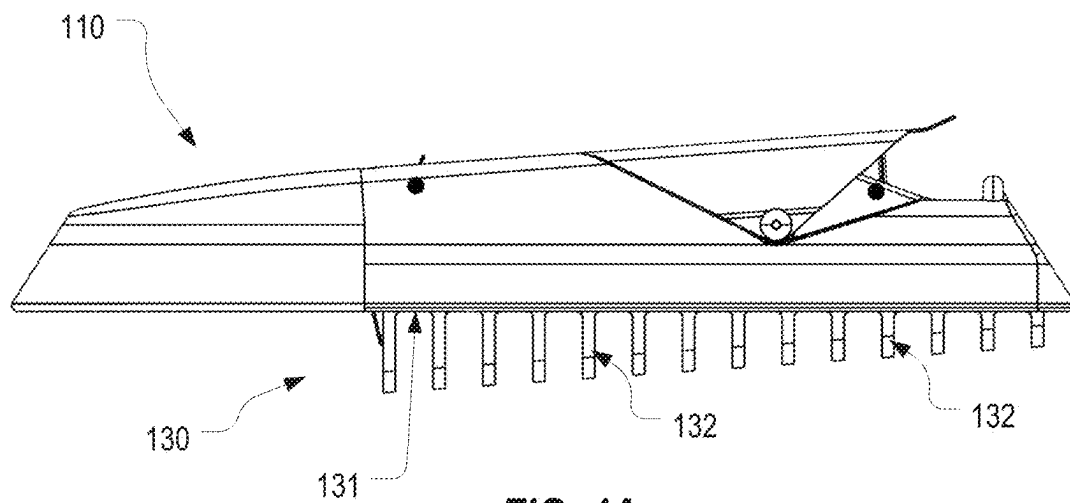


FIG. 4A

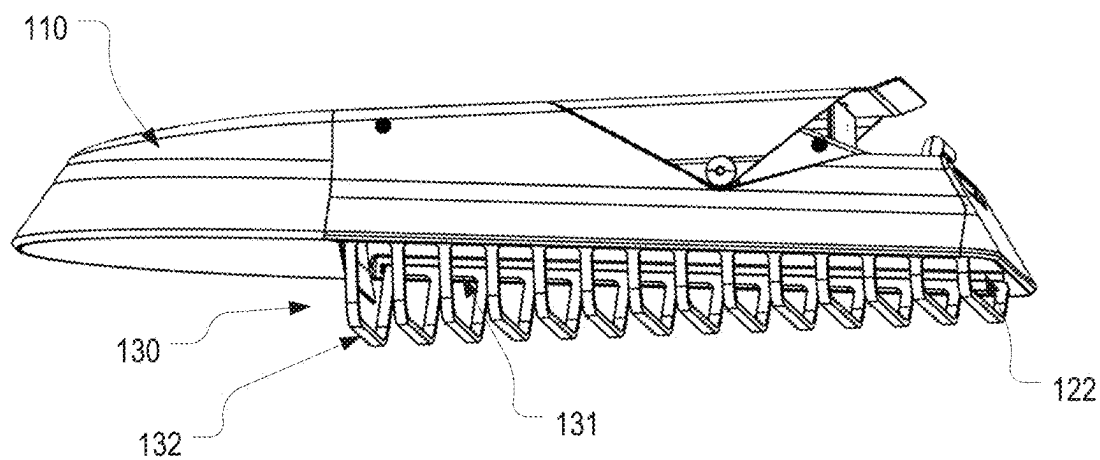
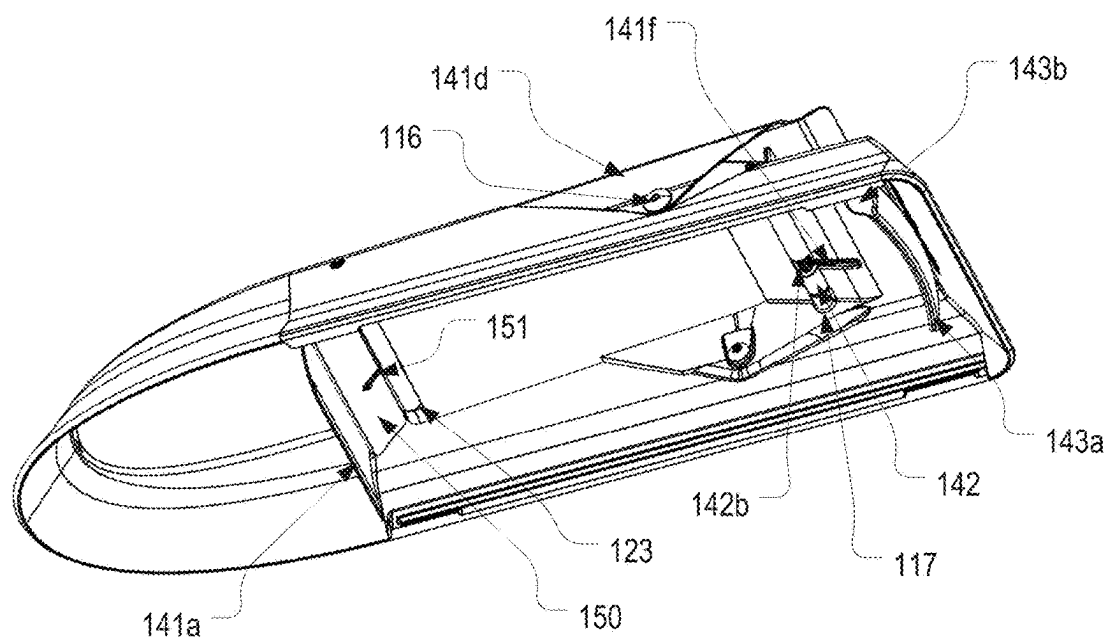
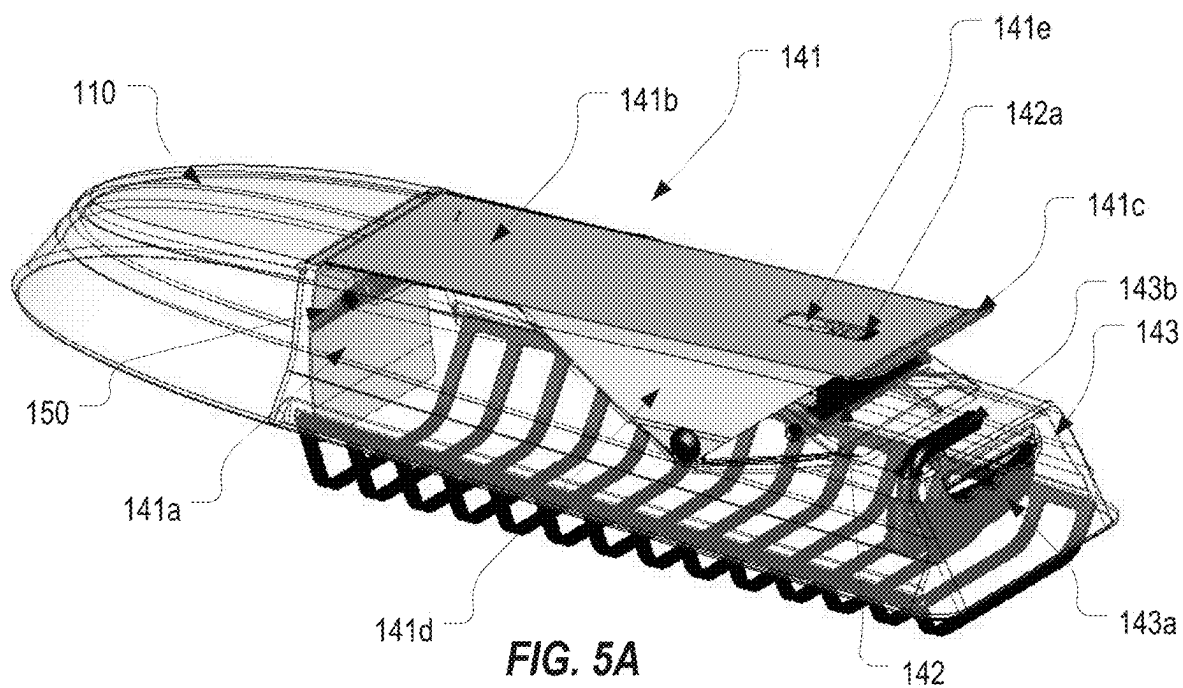


FIG. 4B



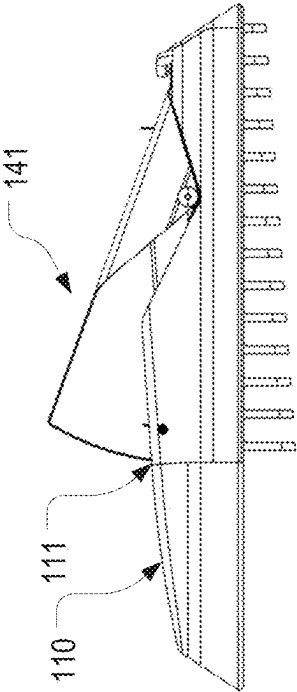


FIG. 6A

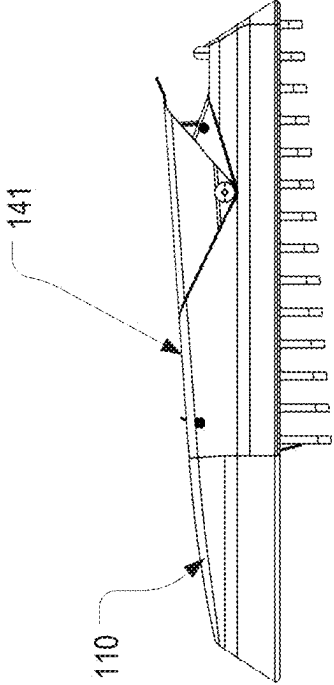


FIG. 6B

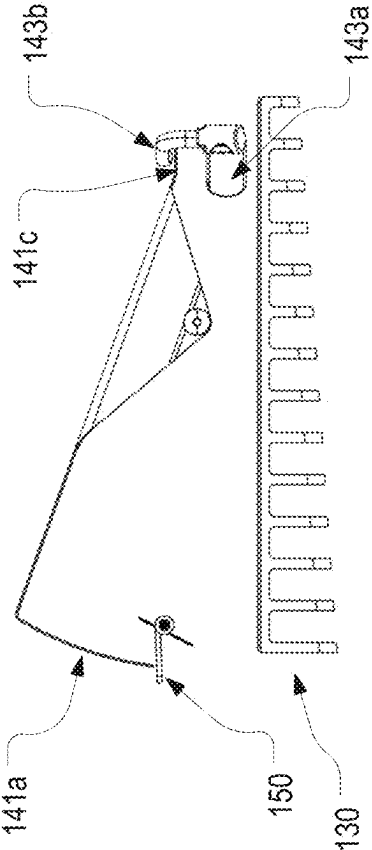


FIG. 7A

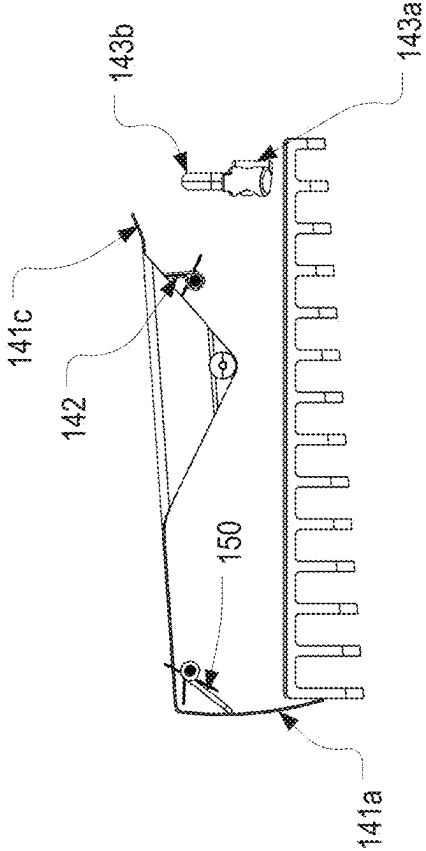


FIG. 7B

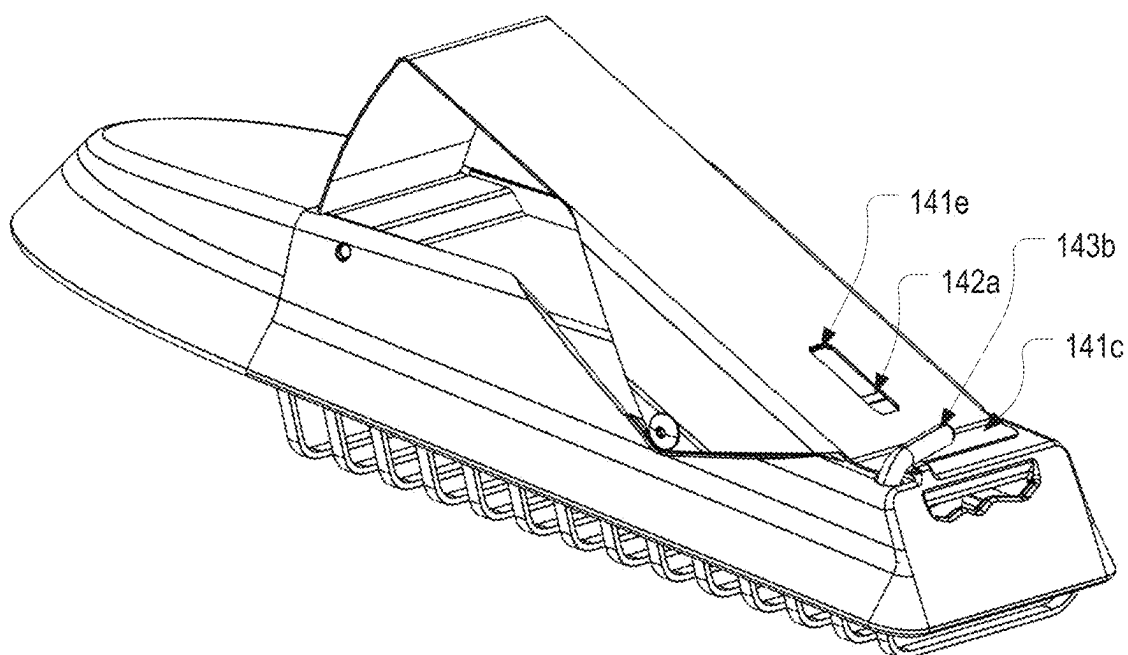


FIG. 8

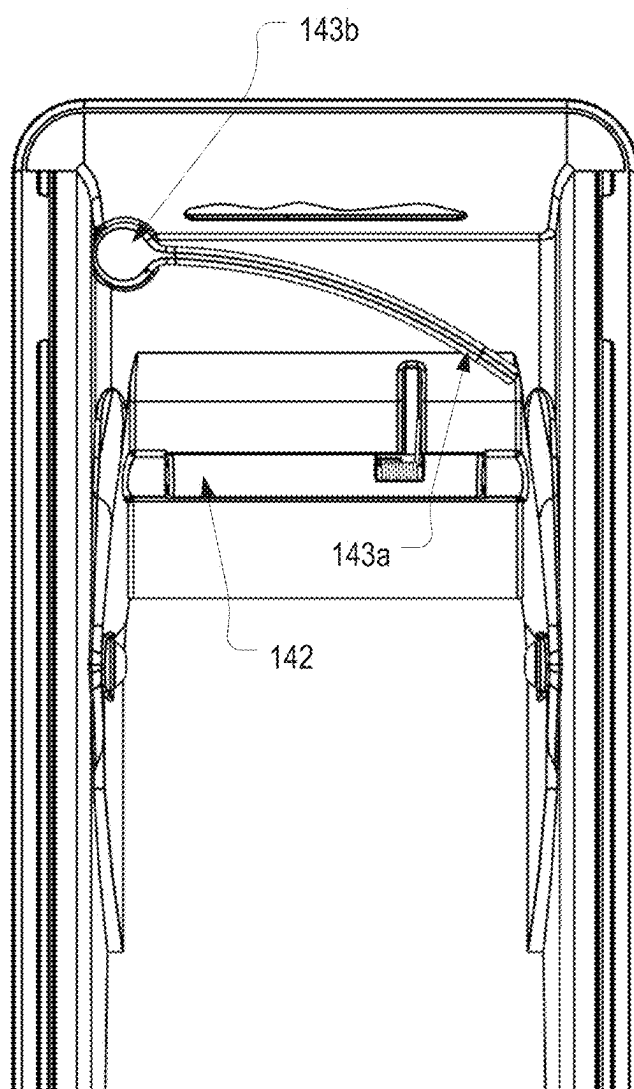


FIG. 9

BURROWING ANIMAL TRAPS

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims the benefit of and incorporates by reference U.S. Provisional Patent Application No. 63/563,552 which was filed on Mar. 11, 2024.

BACKGROUND

[0002] For responsible property management, many jurisdictions require property owners and institutions to control specified animal pests. Even without compulsion, property owners may desire to rid their properties of unwanted pest species whether for their own reasons or as a courtesy to neighbors. Animal pests may be primarily an aesthetic concern but may also present health and safety concerns.

[0003] Different solutions exist for control animal pests including, for example, devices that repel, harm, kill or capture the animals. While it is generally accepted as a positive goal to avoid unnecessary harm to animals, harming or killing animals is at times seen as a necessary solution. Otherwise, repelling animals may work for some species but for others a more effective solution may be capture. Once captured, animals may be released to the wild in a different location or may be otherwise disposed of.

[0004] Animal traps intended for live capture exist but have limited effectiveness on burrowing animals. Many conventional live animal traps rely on bait to attract the animals into an enclosure. However, due to the unfamiliar appearance of such traps, including for example an unnatural floor, the animals are often frightened away. Additionally, attracting animals into an enclosure can take time and is dependent on the animal's appetite and preference for the bait. Moreover, the bait may attract unintended species which may trigger the trap thereby preventing it from capturing the intended species until the trap is reset. These unintended species may even be harmed or killed by the trap.

[0005] Even when a trap successfully captures an animal, additional problems exist. For example, many individuals may be scared of or repulsed by the captured animal. Also, many traps have gaps that may allow the captured animal to bite an individual while handling the trap or may expose the individual to unwanted contact with the captured animal. Furthermore, traps may be difficult or dangerous to set. These additional problems may discourage individuals from employing otherwise effective traps.

[0006] When animals are captured, there is still a risk that the captured animals may be harmed (e.g., due to wind, sun, rain, extreme temperatures, other predatory animals, etc.) before they can be removed from the traps. Also, when a vehicle is used to transfer captured animals to other locations for release, the vehicle will likely become contaminated with soil or feces.

[0007] U.S. Pat. No. 11,723,355, which is incorporated by reference, discloses a burrowing animal trap that addresses at least some of these issues. The present application describes various improvements.

BRIEF SUMMARY

[0008] Embodiments of the present disclosure extend to animal traps that can be used to capture burrowing animals. An animal trap configured in accordance with embodiments

of the present disclosure is designed to enclose the entrance to a burrow and appear to the animal as an extension of the burrow. As a result, the animal will be likely to enter the animal trap under the mistaken belief that it is exiting the burrow.

[0009] In some embodiments, a burrowing animal trap may include an enclosure, a cage floor, a capture mechanism and a trigger mechanism. The enclosure may have a downward-facing opening that is configured to be placed overtop an opening of a burrow. The cage floor may enclose a rear portion of the downward-facing opening to thereby form a frontward-facing opening between the enclosure and the cage floor. The capture mechanism may be configured to selectively cover the frontward-facing opening. The capture mechanism may include a trap door having a top that is connected to the enclosure via a pivoting connection, a front wall that extends downwardly from the top and that is configured to extend into the enclosure to block the frontward-facing opening, and a rear. The front wall and rear may be positioned on opposite sides of the pivoting connection. The trigger mechanism may have a trigger that is positioned within the enclosure and a locking member that rotates with the trigger. The locking member may be configured to interface with the rear of the trap door to retain the front wall in a lifted position until the trigger is rotated.

[0010] In some embodiments, the locking member may be configured to rotate overtop the rear of the trap door to retain the front wall in the lifted position.

[0011] In some embodiments, the rear may be angled upwardly relative to the top of the trap door.

[0012] In some embodiments, the trap door may include opposing sides that extend downwardly from the top. The opposing sides may be connected to the enclosure to form the pivoting connection.

[0013] In some embodiments, the front wall of the trap door may have a curved shape.

[0014] In some embodiments, the capture mechanism may include a biasing mechanism that biases the trap door to cause the front wall to extend into the enclosure.

[0015] In some embodiments, the biasing mechanism may be positioned below the top of the trap door on a rear side of the pivoting connection.

[0016] In some embodiments, the biasing mechanism may rotate rearwardly into a vertical orientation under the trap door.

[0017] In some embodiments, the top of the trap door may include an opening by which the biasing mechanism may be accessed to rotate the biasing mechanism frontward.

[0018] In some embodiments, the enclosure may include a slot through which the front wall of the trap door extends into the enclosure. The burrowing animal trap may further include a slot cover that covers that slot when the front wall is withdrawn from the slot.

[0019] In some embodiments, the slot cover may be configured to pivot away from the slot when the front wall extends through the slot and into the enclosure.

[0020] In some embodiments, the cage floor may include cross members that extend across the enclosure. Each cross member may have vertical portions and a horizontal portion that extends between the vertical portions. A height of the vertical portions may increase in a frontward direction.

[0021] In some embodiments, the height of the vertical portions of the cross members may cause the cross members

to extend along a first plane and a top of the enclosure and the top of the trap door may extend along a second plane that is parallel to the first plane.

[0022] In some embodiments, the burrowing animal trap may include a bottom cover that selectively couples to the enclosure overtop the downward-facing opening. A bottom of the bottom cover may extend along a plane that is parallel to the first and second planes.

[0023] In some embodiments, a burrowing animal trap may include an enclosure, a cage floor, a capture mechanism and a trigger mechanism. The enclosure may have a downward-facing opening that is configured to be placed overtop an opening of a burrow. The enclosure may also have a slot formed on a top of the enclosure. The cage floor may enclose a rear portion of the downward-facing opening to thereby form a frontward-facing opening between the enclosure and the cage floor. The capture mechanism may have a trap door and a biasing mechanism. The trap door may have a top that is connected to the enclosure via a pivoting connection, a front wall that extends downwardly from the top and that is configured to extend through the slot in the enclosure to block the frontward-facing opening, and a rear. The front wall and rear may be positioned on opposite sides of the pivoting connection. The biasing mechanism may apply an upward biasing force on a portion of the trap door that is rearward of the pivoting connection to thereby bias the front wall downwardly through the slot. The trigger mechanism may have a trigger that is positioned within the enclosure and a locking member that interfaces with the trap door to oppose the biasing force until the trigger is moved.

[0024] In some embodiments, the biasing mechanism may be connected to the enclosure via a pivoting connection.

[0025] In some embodiments, the biasing mechanism may include a tab that is accessible through an opening in the top of the trap door to thereby allow a user to apply a frontward force on the tab.

[0026] In some embodiments, the downward-facing opening may extend along a first plane and the cage floor may extend along a second plane. A distance between the first and second plane may increase from a rear of the enclosure to the frontward-facing opening.

[0027] In some embodiments, a burrowing animal trap may include an enclosure having a downward-facing opening that is configured to be placed overtop an opening of a burrow, a cage floor that encloses a rear portion of the downward-facing opening to thereby form a frontward-facing opening between the enclosure and the cage floor, a capture mechanism comprising a trap door having a top that is connected to the enclosure via a pivoting connection and a front wall that extends downwardly from the top and pivots downwardly into the enclosure to block the frontward-facing opening, and a trigger mechanism having a trigger that is positioned within the enclosure and a locking member that rotates with the trigger. The locking member may be configured to interface with the trap door to retain the front wall in a lifted position until the trigger is rotated.

[0028] This summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter.

BRIEF DESCRIPTION OF THE DRAWINGS

[0029] The present disclosure includes the following drawings which represent example and non-limiting embodiments:

[0030] FIGS. 1A and 1B illustrate an example burrowing animal trap that is configured in accordance with one or more embodiments;

[0031] FIGS. 2A and 2B illustrate the example burrowing animal trap with a bottom cover removed;

[0032] FIGS. 3A and 3B illustrate an enclosure of the example burrowing animal trap;

[0033] FIGS. 4A and 4B are side views of the example burrowing animal trap which illustrate a cage floor;

[0034] FIGS. 5A and 5B illustrate a capture mechanism of the example burrowing animal trap;

[0035] FIGS. 6A, 6B, 7A and 7B illustrate how the capture mechanism functions in one or more embodiments;

[0036] FIG. 8 provides another example of the capture mechanism when set; and

[0037] FIG. 9 is a detailed view of the trigger mechanism of the example burrowing animal trap when the capture mechanism is set.

DETAILED DESCRIPTION

[0038] FIGS. 1A and 1B illustrate an example burrowing animal trap **50** that is configured in accordance with one or more embodiments of the present disclosure. Burrowing animal trap **50** has a trap body **100** and a bottom cover **200** that is selectively secured to trap body **100**.

[0039] FIGS. 2A and 2B show trap body **100** with bottom cover **200** removed. Trap body **100** includes an enclosure **110** having a bottom **110a** forming a downward-facing opening, a cage floor **130** that encloses a rear portion of the downward-facing opening, a capture mechanism **140** that is configured to selectively cover a frontward-facing opening formed between enclosure **110** and cage floor **130**, and a slot cover **150** for blocking light from entering enclosure **110**.

[0040] FIGS. 3A and 3B show enclosure **110** in isolation. FIGS. 4A and 4B are side views of burrowing animal trap **50** and provide details of cage floor **130**. FIG. 5A, which shows enclosure **110** with transparency, and FIG. 5B, in which cage floor **130** is removed, provide details of capture mechanism **140**.

[0041] Capture mechanism **140** includes a trap door **141**, a biasing mechanism **142** and a trigger mechanism **143**. Trap door **141** may include a front wall **141a**, a top **141b**, a rear **141c**, sides **141d**, an opening **141e** in top **141b**, and a slot **141f**. Front wall **141a** extends downwardly from a front of top **141b**, whereas rear **141c** forms an upwardly angled portion at the rear of top **141b**. Sides **141d** extend downwardly from the sides of top **141b** and may have triangular shapes, the apexes of which may be secured to enclosure **110** via a pivoting connection. Accordingly, trap door **141** is configured to pivot relative to enclosure **110** as further described below.

[0042] Biasing mechanism **142** is a pivoting structure positioned below top **141b** of trap door **141** and includes a biasing member **142b** (e.g., a spring) that biases the pivoting structure into a vertical position (e.g., as shown in FIG. 5A). Biasing mechanism **142** can include a tab **142a** that aligns with opening **141e** in top **141b**. Because biasing mechanism **142** is positioned to the rear of trap door **141**'s pivoting

connection, biasing mechanism **142** applies an upward biasing force on the rear of trap door **141** such that front wall **141a** is biased downwardly.

[0043] Trigger mechanism **143** includes a trigger **143a** and a locking member **143b**. As described in greater detail below, trigger **143a** is positioned within enclosure where an animal or soil may contact it. Locking member **143b** is configured to selectively secure rear **141c** to retain trap door **141** in a set position.

[0044] Enclosure **110** includes a slot **111** through which front wall **141a** inserts into enclosure **110** and an opening **112** through which biasing mechanism **142** extends out from enclosure **110**. In some embodiments, opening **112** may be formed in/along a recessed portion **113** of enclosure **110**'s exterior. In some embodiments, the exterior of enclosure **110** may form stops **114** that slides **141d** contact to thereby limit the movement of trap door **141**. In some embodiments, the exterior of enclosure **110** may form a ledge **115** along and behind slot **111**. Top **141b** may rest on ledge **115** when front wall **141a** is fully inserted through slot **111**. In some embodiments, enclosure **110** may form pivot points (e.g., openings) **116** for trap door **141** and pivot points **117** for biasing mechanism **142**. In some embodiments, enclosure **110** may form an opening **118** for trigger mechanism **143**. In some embodiments, enclosure **110** may include an opening **119** for sunlight which is positioned at a rear of the enclosure. Opening **119** can create the illusion that enclosure **110** is an extension of the burrow. In some embodiments, enclosure **110** may form a support **120** for trigger mechanism **143**. In some embodiments, enclosure **110** may include a channel **121** for biasing member **142b**. In some embodiments, enclosure **110** may form a retaining structure **122** for cage floor **130**. In some embodiments, an interior of enclosure **110** may form pivot points **123** for slot cover **150**.

[0045] Cage floor **130** includes side members **131** and cross members **132** which, in some embodiments, may be formed from stamped metal, welded wire, or another suitable configuration. Side members **131** are configured to interface with retaining structure **122** to secure cage floor **130** to enclosure. Cross members **132** extend between side members **131** and may span from the rear of enclosure **110** up to the location of front wall **141a** when front wall **141a** is inserted into enclosure **110**. Each cross member **132** may include opposing vertical portions and a horizontal portion that extends between the vertical portions. As best seen in FIG. 4A, the vertical portions of cross members **132** can be configured so that their height increases in the forward direction. In other words, the horizontal portions of cross members **132** positioned at and near front wall **141a** are spaced at a greater distance from bottom **110a** than those positioned at and near the rear of enclosure **110**. By configuring cage floor **130** with this sloped design, cross members **132** are less likely to be exposed above the soil at the entrance into enclosure **110** and can therefore minimize the likelihood that an animal will detect the trap. Bottom cover **200** can be shaped to generally match this sloped shape of cage floor **130**. To facilitate stacking of traps **50**, the top contour of enclosure **110** (and trap door **141** when closed) can be configured to run parallel with the sloped shape of cage floor **130**.

[0046] FIG. 5A shows enclosure **110** with transparency and FIG. 5B omits cage floor **130** to better illustrate how the various components interconnect and function. As shown, trap door **141** is coupled to enclosure **110** by sides **141d** and

via a pivoting connection. Front wall **141a** descends into enclosure **110** via slot **111** to close the front-facing opening between enclosure **110** and cage floor **130**. Biasing mechanism **142** biases trap door **141** into this closed position and also prevents pivoting of trap door **141** once in the closed position.

[0047] Slot cover **150** is configured to cover slot **111** when front wall **141a** is lifted to thereby block light from entering through slot **111** and maintain the illusion that an animal is still in the burrow when entering enclosure **110**. Slot cover **150** includes a biasing member **151** that biases slot cover **150** overtop slot **111** but also allows slot cover **150** to pivot downwardly out of the way as front wall **141a** descends into enclosure **110**.

[0048] Trigger **143a** is positioned towards the rear of enclosure **110** and is configured to pivot between a set (forward) and triggered (rearward) position. Locking member **143b** is configured to pivot with trigger **143a**. Accordingly, by rotating locking member **143b** overtop rear **141c** of trap door **141**, trigger **143a** will also be rotating forward into its set position. Then, an animal or soil the animal may move can force trigger **143a** rearward to thereby rotate locking member **143b** off of rear **141c**. Biasing member **142b**, with the help of gravity, will then rotate trap door **141** causing front wall **141a** to descend into and be held in the closed position to trap the animal within enclosure **110**.

[0049] FIGS. 6A and 6B are side views of burrowing animal trap **50** in the set and triggered positions respectively. FIGS. 7A and 7B are equivalent views of capture mechanism **140** in isolation. FIG. 8 is a rear view of burrowing animal trap **50** when set. In the bottom view of FIG. 9, cage floor **130** is removed to show trigger **143a** when set. These figures further detail how burrowing animal trap **50** can be configured to function in some embodiments.

[0050] With reference to FIG. 7B, when burrowing animal trap **50** is in the triggered (or unset) position, biasing mechanism **142** is in an upward orientation and prevents downward movement of the rear portion of trap door **141** which thereby prevents upward movement of front wall **141a**. To transition burrowing animal trap **50** from the triggered position to the set position (i.e., to transition from FIG. 7B to FIG. 7A), the user can push tab **142a** forward via opening **141e** while pushing downwardly on rear **141c** to thereby pivot front wall **141a** upwardly out from slot **111**. As front wall **141a** exits slot **111**, slot cover **150** is biased overtop slot **111** to block light. Then, with rear **141c** lowered, the user can rotate locking member **143b** overtop rear **141c** to retain trap door **141** in the set position as shown in FIG. 8. This rotation of locking member **143b** also rotates trigger **143a** forwardly as shown in FIG. 9. Then, when an animal enters enclosure **110**, it can contact trigger **143a** and push it rearward which in turn rotates locking member **143b** off of rear **141c**. With locking member **143b** no longer holding rear **141c**, biasing member **142**, in conjunction with gravity, causes trap door **141** to pivot back to the triggered position.

[0051] Burrowing animal trap **50** is designed to be more reliable even with the small forces that animals may apply. For example, all the components of capture mechanism **140** are connected to enclosure **110** via a pivoting connection. Such pivoting connections minimize friction (particularly in comparison to sliding components) which can cause capture mechanism **140** to be triggered by even a small force against trigger **143a**.

What is claimed:

1. A burrowing animal trap comprising:
 - an enclosure having a downward-facing opening that is configured to be placed overtop an opening of a burrow;
 - a cage floor that encloses a rear portion of the downward-facing opening to thereby form a frontward-facing opening between the enclosure and the cage floor;
 - a capture mechanism that is configured to selectively cover the frontward-facing opening, the capture mechanism comprising a trap door having a top that is connected to the enclosure via a pivoting connection, a front wall that extends downwardly from the top and that is configured to extend into the enclosure to block the frontward-facing opening, and a rear, the front wall and rear being positioned on opposite sides of the pivoting connection; and
 - a trigger mechanism having a trigger that is positioned within the enclosure and a locking member that rotates with the trigger, wherein the locking member is configured to interface with the rear of the trap door to retain the front wall in a lifted position until the trigger is rotated.
2. The burrowing animal trap of claim 1, wherein the locking member is configured to rotate overtop the rear of the trap door to retain the front wall in the lifted position.
3. The burrowing animal trap of claim 2, wherein the rear is angled upwardly relative to the top of the trap door.
4. The burrowing animal trap of claim 1, wherein the trap door includes opposing sides that extend downwardly from the top, the opposing sides being connected to the enclosure to form the pivoting connection.
5. The burrowing animal trap of claim 1, wherein the front wall of the trap door has a curved shape.
6. The burrowing animal trap of claim 1, wherein the capture mechanism includes a biasing mechanism that biases the trap door to cause the front wall to extend into the enclosure.
7. The burrowing animal trap of claim 6, wherein the biasing mechanism is positioned below the top of the trap door on a rear side of the pivoting connection.
8. The burrowing animal trap of claim 7, wherein the biasing mechanism rotates rearwardly into a vertical orientation under the trap door.
9. The burrowing animal trap of claim 8, wherein the top of the trap door includes an opening by which the biasing mechanism may be accessed to rotate the biasing mechanism frontward.
10. The burrowing animal trap of claim 1, wherein the enclosure includes a slot through which the front wall of the trap door extends into the enclosure, the burrowing animal trap further comprising:
 - a slot cover that covers that slot when the front wall is withdrawn from the slot.
11. The burrowing animal trap of claim 10, wherein the slot cover is configured to pivot away from the slot when the front wall extends through the slot and into the enclosure.
12. The burrowing animal trap of claim 1, wherein the cage floor comprises cross members that extend across the enclosure, each cross member having vertical portions and a horizontal portion that extends between the vertical portions, wherein a height of the vertical portions increases in a frontward direction.
13. The burrowing animal trap of claim 12, wherein the height of the vertical portions of the cross members cause the cross members to extend along a first plane and wherein a top of the enclosure and the top of the trap door extend along a second plane that is parallel to the first plane.
14. The burrowing animal trap of claim 13, further comprising:
 - a bottom cover that selectively couples to the enclosure overtop the downward-facing opening, wherein a bottom of the bottom cover extends along a plane that is parallel to the first and second planes.
15. A burrowing animal trap comprising:
 - an enclosure having a downward-facing opening that is configured to be placed overtop an opening of a burrow, the enclosure further comprising a slot formed on a top of the enclosure;
 - a cage floor that encloses a rear portion of the downward-facing opening to thereby form a frontward-facing opening between the enclosure and the cage floor;
 - a capture mechanism comprising a trap door and a biasing mechanism, the trap door having a top that is connected to the enclosure via a pivoting connection, a front wall that extends downwardly from the top and that is configured to extend through the slot in the enclosure to block the frontward-facing opening, and a rear, the front wall and rear being positioned on opposite sides of the pivoting connection, the biasing mechanism applying an upward biasing force on a portion of the trap door that is rearward of the pivoting connection to thereby bias the front wall downwardly through the slot; and
 - a trigger mechanism having a trigger that is positioned within the enclosure and a locking member that interfaces with the trap door to oppose the biasing force until the trigger is moved.
16. The burrowing animal trap of claim 15, wherein the biasing mechanism is connected to the enclosure via a pivoting connection.
17. The burrowing animal trap of claim 16, wherein the biasing mechanism includes a tab that is accessible through an opening in the top of the trap door to thereby allow a user to apply a frontward force on the tab.
18. The burrowing animal trap of claim 15, wherein the downward-facing opening extends along a first plane and the cage floor extends along a second plane, wherein a distance between the first and second plane increases from a rear of the enclosure to the frontward-facing opening.
19. A burrowing animal trap comprising:
 - an enclosure having a downward-facing opening that is configured to be placed overtop an opening of a burrow;
 - a cage floor that encloses a rear portion of the downward-facing opening to thereby form a frontward-facing opening between the enclosure and the cage floor;
 - a capture mechanism comprising a trap door having a top that is connected to the enclosure via a pivoting connection and a front wall that extends downwardly from the top and pivots downwardly into the enclosure to block the frontward-facing opening; and
 - a trigger mechanism having a trigger that is positioned within the enclosure and a locking member that rotates with the trigger, wherein the locking member is configured to interface with the trap door to retain the front wall in a lifted position until the trigger is rotated.

20. The burrowing animal trap of claim **19**, wherein the cage floor comprises cross members that extend across the enclosure, each cross member having vertical portions and a horizontal portion that extends between the vertical portions, wherein a height of the vertical portions increases in a frontward direction.

* * * * *